

IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

EXAMINATIONS 2023

BEng Honours Degree in Computing Part I
MEng Honours Degrees in Computing Part I
for Internal Students of the Imperial College of Science, Technology and Medicine

*This paper is also taken for the relevant examinations for the
Associateship of the City and Guilds of London Institute*

PAPER COMP40018A

DISCRETE MATHEMATICS, LOGIC AND REASONING (PART 1)

Thursday 4th May 2023, 14:00
Duration: 80 minutes

Answer ALL TWO questions

Paper contains 2 questions
Calculators not required

- 1 a Let $V = \{a, b, \{b\}\}$ and $W = \{a, \{a\}, b\}$. Determine $V \cap W$, $V \cup W$, $V \setminus \wp W$, and $\wp V \cup W$.
- b Show that $(\overline{R \cup S})^{-1} = \overline{R^{-1}} \cap \overline{S^{-1}}$.
- You do not need to give a formal proof, and can use the abbreviated notation for statements. Write clearly when a step follows by definition, or when by a logical reasoning, or when information from the notes is used.
- c Let $\mathcal{V} \subseteq \wp \mathbb{N}$ be defined as $\mathcal{V} \triangleq \{V_n \mid n \in \mathbb{N}\}$, where each V_n is defined as $\{m \in \mathbb{N} \mid \exists k \in \mathbb{N} (n \times k = m)\}$. Does $\langle \mathcal{V}, \subseteq \rangle$ have
- minimal elements,
 - least elements,
 - maximal elements, or
 - greatest elements?
- In each case, specify whether such an element is unique, if it exists? If it does not exist, argue why. Give examples, and show why each satisfies the criterion.
- d Determine which of the following relations are (partial) functions. For those which are functions, determine whether they are injective and/or surjective. Also, give the inverse function when it exists. Justify your answer throughout.
- $\{\langle x, y \rangle \in \mathbb{Z}^2 \mid x = y \wedge x = -y\}$;
 - $\{\langle x, y \rangle \in (\mathbb{R} \setminus \{2\}) \times (\mathbb{R} \setminus \{0\}) \mid y = 1/(x^3 - 8)\}$;
 - $\{\langle x, y \rangle \in \mathbb{R}^2 \mid x^2 = y^3\}$.
 - $\{\langle x, y \rangle \in \mathbb{Q}^2 \mid y = x^3\}$.
- e For your answers below, clearly state any result from the course that you may use in your proof; your proof should not use the ‘by contradiction’ step.
- Show that the set of all functions from $\mathbb{N}$ to $\{1\}$ is countable.
 - Show that the set of all partial functions from $\mathbb{N}$ to $\{1\}$ is not countable.

The five parts carry equal marks.

- 2 a i) The set $\{\vee, \neg\}$ is functionally complete. Using truth tables, show that the set $\{\leftrightarrow, \wedge, \perp\}$ is also functionally complete.

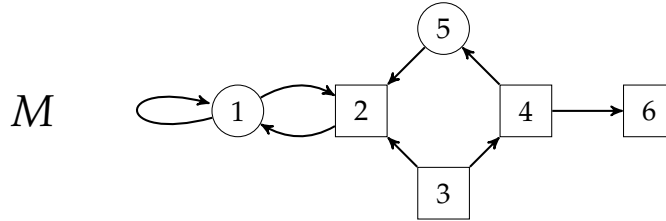
- ii) Using direct argument, determine whether the following formula is a valid formula.

$$(((p \rightarrow q) \rightarrow \neg q) \rightarrow \neg q)$$

- iii) Rewrite the following propositional formula into conjunctive normal form. Specify at each step the transformation rule applied.

$$(p \wedge q) \rightarrow \neg(r \rightarrow \neg \neg s)$$

- b i) Let L be a signature consisting of a unary relation symbol C and a binary relation symbol S (read $S(x, y)$ as “ x sees y ”). Below is a diagram of an L -structure M . The arrows indicate the interpretation of S in M , and the round nodes are the objects satisfying C , e.g., $M \models S(5, 2) \wedge \neg S(2, 5) \wedge C(1)$.



- A) The L -structure M shows that the formula $C(x)$ is true in M only for $x = 1$ and $x = 5$. Similarly, list all objects of x in $dom(M)$ for which the following formulae are true in M . Explain your answer.

- i) $\forall y(S(x, y) \rightarrow \exists z(S(y, z) \wedge S(z, z)))$
- ii) $\exists y \forall z(C(z) \wedge S(z, x) \rightarrow z = y)$
- iii) $\exists y \exists z(y \neq z \wedge \forall w(S(x, w) \leftrightarrow w = y \vee w = z))$

- B) Consider an additional unary relation symbol Q . State what objects in M must be included in the interpretation of Q for the following sentence to be true in M .

$$\forall x(Q(x) \leftrightarrow \neg \exists y(S(x, y) \wedge Q(y)))$$

- ii) Using natural deduction, show that

$$\exists x \forall y \exists z(A(x) \rightarrow B(y, z)) \vdash \forall x A(x) \rightarrow \forall y \exists z(B(y, z) \vee C(z))$$

The two parts carry equal marks.